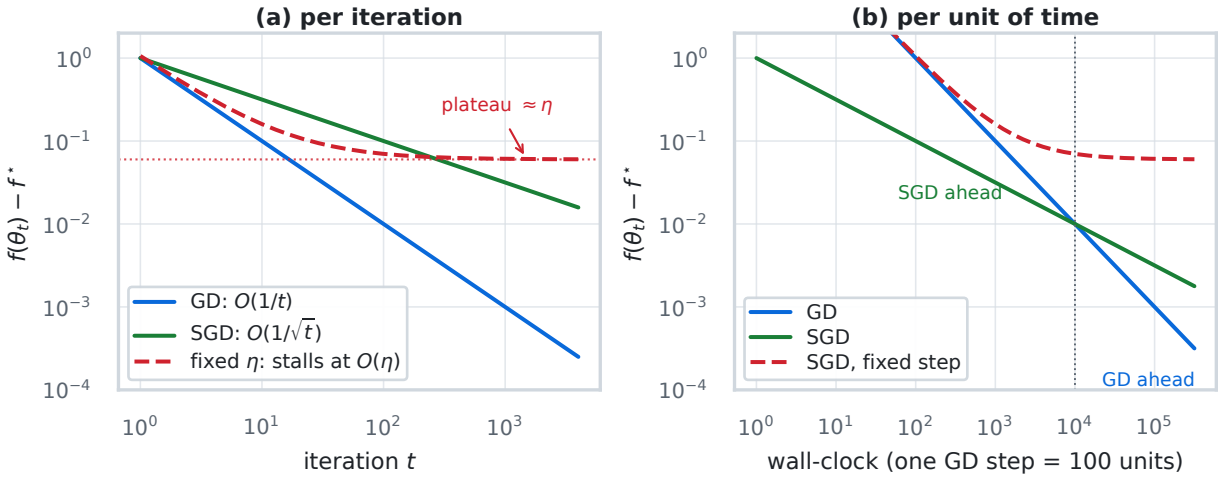
(a) per iteration
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
$f(\theta_t) - f^\star$
plateau $\approx \eta$
GD: $O(1/t)$
SGD: $O(1/\sqrt{t})$
fixed η: stalls at $O(\eta)$
10^0
10^1
10^2
10^3
iteration t
(b) per unit of time
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
$f(\theta_t) - f^\star$
SGD ahead
GD
SGD
SGD, fixed step
GD ahead
10^0
10^1
10^2
10^3
10^4
10^5
wall-clock (one GD step = 100 units)